

IEEE P802.1DG™/D4.2

Draft Standard for Local and metropolitan area networks — Time-Sensitive Networking Profile for Automotive In-Vehicle Ethernet Communications

Prepared by the Time-Sensitive Networking Task Group of IEEE 802.1 of the
LAN MAN Standards Committee
of the
IEEE Computer Society

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18 **Abstract:** This standard specifies profiles for bounded latency automotive in-vehicle Bridged IEEE
19 802.3 Ethernet networks based on IEEE 802.1 Time-Sensitive Networking (TSN) standards.

20 **Keywords:** TSN, Time-Sensitive Networking, Bridging, Bridges, Bridged Local Area Networks,
21 IEEE 802®, IEEE 802.1Q™, IEEE 802.1DG™, local area networks (LANs), MAC Bridges, Virtual
22 Bridged Local Area Networks (virtual LANs), Automotive In-Vehicle Ethernet Communications.

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This introduction is not part of IEEE P802.1DG™/D4.2, IEEE Standards for Local and Metropolitan Area Networks—Draft Standard for Local and metropolitan area networks —Time-Sensitive Networking Profile for Automotive In-Vehicle Ethernet Communications

5 This Standard defines the Time-Sensitive Networking Profile for Automotive In-Vehicle Ethernet
6 Communications.

7 This standard contains state-of-the-art material. The area covered by this standard is undergoing evolution.
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1 IEEE P802.1DG™/D4.2

2 Draft Standard for 3 Local and metropolitan area 4 networks—

5 6 Time-Sensitive Networking Profile for 7 Automotive In-Vehicle Ethernet 8 Communications

9 1. Overview

10 1.1 Scope

11 This standard specifies profiles for bounded latency automotive in-vehicle Bridged IEEE 802.3 Ethernet
12 networks based on IEEE 802.1 Time-Sensitive Networking (TSN) standards.

13 1.2 Purpose

14 This standard provides profiles for designers and implementers of automotive IEEE 802.3 Ethernet
15 networks that support a wide range of in-vehicle applications.

16 1.3 Requirements terminology

17 “Supported” - The capability is available.

18 “Activated” - The capability is available (i.e., it is supported) and enabled.

19 The following terminology is used in Clause 5 and Annex A:

- 20 a) The word “shall” indicates mandatory requirements strictly to be followed in order to conform to the
21 standard and from which no deviation is permitted (“shall” equals “is required to”).
- 22 b) The word “should” indicates that among several possibilities, one is recommended as particularly
23 suitable, without mentioning or excluding others; or that a certain course of action is preferred but
24 not necessarily required (“should” equals “is recommended that”, but not required).
- 25 c) The word “may” is used to indicate a course of action permissible within the limits of the standard
26 (may equals is permitted to).

1 All other clauses than Clause 5 and the clauses of Annex A use the following terminology:

- 2 d) The words “might” or “can” are used for statements of possibility (“might”/“can” mean “there is a
3 possibility that”).
- 4 e) The neutral words “is” or “does” are used to describe a capability, procedure, or behavior;
5 independent of it being required, prohibited, recommended, discouraged, or permitted.
- 6 f) “Not activated” - The capability is available but not enabled.

7 **1.4 Introduction**

8 This Time-Sensitive Networking (TSN) Profile for Automotive In-Vehicle Ethernet Communications
9 standard addresses the use of features from IEEE 802.1 standards to meet the bandwidth, latency, and
10 synchronization needs for communications within automotive vehicles. The profile introduces the subtleties
11 of the operation of the IEEE 802.1 standards and the side-effects of the choices made when configuring
12 various features from IEEE 802.1 standards. The goal of this standard is to provide information to
13 automotive vendors and suppliers to help them with the design of vehicular systems enabling bounded
14 latency in automotive in-vehicle networks. As IEEE 802.1 standards are broad and intended for use in a
15 variety of environments, this standard narrows the focus from the broad set of standardized features to those
16 that are applicable to in-vehicle networks (IVN). This standard determines the features from IEEE 802.1
17 standards that are directly applicable to IVNs and explains how these features are used, including
18 recommendations about how to configure optional parameters.

19 **1.5 Outline of the document structure**

20 This standard specifies bridge and end station requirements and options in Clause 5. All conformance
21 language is aggregated in Clause 5. Clauses 6. through 8. describe the TSN features in more detail and
22 provide references to the base standards.

1 1.6 Reference conventions

2 Because this standard makes frequent references to specific sections, clauses, or subclauses in several other
3 standards and amendments, it uses the notations described in Table 1-1.

Table 1-1—Conventions for references

Reference shorthand notation	Complete reference
[AC] x.y	subclause x.y in IEEE Std 802.1AC-2016
[AE] x.y	subclause x.y in IEEE Std 802.1AE-2018 ([B3])
[BA] x.y	subclause x.y in IEEE Std 802.1BA-2021 ([B4])
[CB] x.y	subclause x.y in IEEE Std 802.1CB-2017
[CBdb] x.y	subclause x.y in IEEE Std 802.1CBdb-2021
[DC] x.y	subclause x.y in IEEE Std 802.1DC-2024
[Q] x.y	subclause x.y in IEEE Std 802.1Q-2022
[B##] x.y	section or subclause x.y in any document from the Bibliography in Annex I
[802] x.y	subclause x.y in IEEE Std 802-2014
[802.3]x.y	subclause x.y in IEEE Std 802.3-2022
x.y	subclause x.y in this standard

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NOTE 1—The inclusion of a document in this list of normative references indicates that information in that document is necessary to implement this standard. It does not imply that any other part of that referenced document is required to be implemented by a system conformant to this standard.

IEEE Std 802.1AC™-2016, IEEE Standard for Local and metropolitan area networks—Media Access Control (MAC) Service Definition.

IEEE Std 802.1CB™-2017, IEEE Standard for Local and metropolitan area networks—Frame Replication and Elimination for Reliability.

IEEE Std 802.1CBdb™-2021, IEEE Standard for Local and metropolitan area networks—Frame Replication and Elimination for Reliability. Amendment 2: Extended Stream Identification Functions.

IEEE Std 802.1DC™-2024, IEEE Standard for Local and Metropolitan Area Networks—Quality of Service Provision by Network Systems.

IEEE Std 802.1Q™-2022, IEEE Standard for Local and metropolitan area networks—Bridges and Bridged Networks.

IEEE Std 802[®]-2014, IEEE Standard for Local and Metropolitan Area Networks: Overview and Architecture.^{1,2,3}

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This standard makes use of the following terms defined in IEEE Std 802 ([802]):

- bridge
- end station
- station

This standard makes use of the following terms defined in IEEE Std 802.3 ([802.3]):

- frame
- packet

This standard makes use of the following terms defined in IEEE Std 802.1Q ([Q]):

- C-VLAN

1. The IEEE Standards Dictionary Online is available at: <http://dictionary.ieee.org>.

4. Abbreviations

This standard uses the following abbreviations:

3	ATS	Asynchronous Traffic Shaper
5	BE	Best Effort
7	CBS	Credit Based Shaper
9	CQF	Cyclic Queuing and Forwarding
11	ECU	Electronic Control Unit
13	EISS	Enhanced Internal Sublayer Service
15	EST	Enhancements for Scheduled Traffic
17	FRER	Frame Replication and Elimination for Reliability
19	ICV	Integrity Check Value
21	IVN	In-Vehicle Network
23	IPG	InterPacket Gap
25	IPV	Internal Priority Value
27	ISS	Internal Sublayer Service
29	MSDU	MAC Service Data Unit
31	PCS	Profile Conformance Statements
33	RSTP	Rapid Spanning Tree Protocol
35	TDMA	Time Division Multiple Access
37	SDU	Service Data Unit
39	SFD	Start of Frame Delimiter
41	SotA	Software over the Air update
43	TSN	Time-Sensitive Networking
45	TC	Traffic Class

47

1 5. Conformance

2 5.1 Introduction

3 This clause specifies the mandatory and optional capabilities provided by conformant implementations of
4 this standard.

5 5.2 Profile Conformance Statements (PCS)

6 The supplier of an implementation that is claimed to conform to this standard shall complete a copy of the
7 PCS proforma provided in Annex A and shall provide the information necessary to identify both the supplier
8 and the implementation.

9 5.3 Physical Layer requirements

10 A station claiming compliance with this standard shall support an IEEE Std 802.3 point-to-point full-duplex
11 ([802.3] 1.4.345) link segment ([802.3] 1.4.379).

12 5.4 Bridge requirements

13 5.4.1 Mandatory Bridge requirements

14 A bridge claiming conformance to this standard shall, on all of its ports:

- 15 a) Support C-VLAN tags ([Q] 5.5).
- 16 b) Conform to SR-A as specified in 5.6.
- 17 c) Conform to SR-B as specified in 5.7.
- 18 d) Conform to SR-C as specified in 5.8.
- 19 e) Support the Learning Process as specified in [Q] 8.7. Additional informative explanations are
20 provided in Annex D.14.
- 21 f) Support the maximum SDU size as specified in [Q] 8.6.5.3.1. Additional informative explanations
22 are provided in Annex D.17.2.
- 23 g) Support the Ingress filtering as specified in [Q] 8.6.2. Additional informative explanations are
24 provided in Annex D.15.
- 25 h) Support the Egress filtering as specified in [Q] 8.6.4. Additional informative explanations are
26 provided in Annex D.15.
- 27 i) Support the Frame Filtering as specified in [Q] 8.6.3. Additional informative explanations are
28 provided in Annex D.15.
- 29 j) Support at least 1024 MAC Address Registration Entries in the Filtering Database ([Q] 8.8.4).
- 30 k) Support the Static IPV specification as specified in [Q] 8.6.5.4. Additional informative explanations
31 are provided in Annex D.18.1.
- 32 l) Support Stream Gating as specified in [Q] 8.6.5.4. Additional informative explanations are provided
33 in Annex D.18.

1 5.4.2 Bridge options

2 A bridge claiming conformance to this standard:

- 3 a) Should support at least one Time Synchronization mechanism as stated in Annex D.5.
- 4 b) May conform to SR-D as specified in 5.9.
- 5 c) May conform to SR-E as specified in 5.10.
- 6 d) May support MACsec as specified in [AE] 11.4. Additional informative explanations are provided
- 7 in Annex D.6.

8 5.5 End Station requirements

9 5.5.1 Mandatory End Station ingress requirements

10 An end station claiming conformance to this standard shall:

- 11 a) Conform to SR-B as specified in 5.7.
- 12 b) Support Out-facing Ingress Stream Identification Function(s) as specified in [CB] 9.1.1.5.
- 13 Additional informative explanations are provided in Annex D.11.2.
- 14 c) Support a minimum number of 64 stream identification functions.
- 15 d) Support Stream Filter assignment as specified in [DC] 7.2.1.2. Additional informative explanations
- 16 are provided in Annex D.16.
- 17 e) Support max. SDU Size Filtering as specified in [DC] 7.2.1.2. Additional informative explanations
- 18 are provided in Annex D.17.2.

19 5.5.2 Mandatory End Station egress requirements

20 An end station claiming conformance to this standard shall:

- 21 a) Support the ATS Talker transmission behavior as specified in [Q] 47.1 on one TC.
- 22 b) Support the Credit Based Shaper (CBS) algorithm as specified in [Q] 34.6.1.1 on the two
- 23 numerically highest value TCs.
- 24 c) Support Queuing frames as specified in [DC] 7.2.2. Additional informative explanations are
- 25 provided in Annex D.20.
- 26 d) Support 8 traffic classes (TCs).
- 27 e) Support Transmission selection as specified in [DC] 7.2.4. Additional informative explanations are
- 28 provided in D.21.

29 5.5.3 End Station Options

30 An end station claiming conformance to this standard may:

- 31 a) Support Time Synchronization as stated in Annex D.5.
- 32 b) Conform to SR-D as specified in 5.9.
- 33 c) Conform to SR-E as specified in 5.10.
- 34 d) Support MACsec processing as specified in [AE]. Additional informative explanations are provided
- 35 in Annex D.6.

- 1 e) Support Stream Gating on ingress as specified in [Q] 8.6.5.4. Additional informative explanations
- 2 are provided in Annex D.18.2.
- 3 f) Support Flow metering on ingress as specified in [Q] 8.6.5.5.

4 5.6 Station Requirements A (SR-A)

5 A station claiming conformance to SR-A shall:

- 6 a) Support Port-based VLAN Classification as specified in item f) of [Q] 6.9.1. Additional informative
- 7 explanations are provided in Annex D.8.
- 8 b) Support the Support Priority Code Point Decoding as specified in [Q] 6.9.3. Additional informative
- 9 explanations are provided in Annex D.9.
- 10 c) Support the Priority Regeneration as specified in [Q] 6.9.4. Additional informative explanations are
- 11 provided in Annex D.10.
- 12 d) Support the Ingress Stream Identification Function(s) as specified in [CB] 6.2. Additional
- 13 informative explanations are provided in Annex D.11.1.
- 14 e) Support the following number of stream identification functions:
- 15 1) 128 identification functions for bridges with less than 5 ports,
- 16 2) 192 identification functions for bridges with between 5 and 9 ports,
- 17 3) 256 identification functions for bridges with more than 9 ports.
- 18 f) Support the Active topology enforcement as specified in [Q] 8.6.1. Additional informative
- 19 explanations are provided in Annex D.12.
- 20 g) Support the Ingress filtering as specified in [Q] 8.6.2. Additional informative explanations are
- 21 provided in Annex D.13.
- 22 h) Support Egress Filtering as specified in [Q] 8.6.4.
- 23 i) Support Queuing frames as specified in [Q] 8.6.6. Additional informative explanations are provided
- 24 in Annex D.20.
- 25 j) Support 8 traffic classes (TCs).
- 26 k) Support Transmission selection as specified in [Q] 8.6.8. Additional informative explanations are
- 27 provided in Annex D.21.
- 28 l) Support the ATS transmission selection algorithm as specified in [Q] 47.1 on one TC.
- 29 m) Support the Credit Based Shaper (CBS) algorithm as specified in [Q] 34.6.1.1 on the two
- 30 numerically highest value TCs.
- 31 n) Support the VID Translation and Egress VID Translation as specified in item g) of [Q] 6.9.
- 32 Additional informative explanations are provided Annex D.23.

33 5.7 Station Requirements B (SR-B)

34 A station claiming conformance to SR-B shall:

- 35 a) Support the Acceptable Frame Types as specified in item c) of [Q] 6.9 for bridges, and item b)1) in
- 36 [DC] 7.3.2 for end stations. Additional informative explanations are provided in Annex D.7.
- 37 b) Support the Maximum SDU Size Filtering as specified in [Q] 8.6.5.3.1 for bridges, and item c) in
- 38 [DC] 7.2.1.2 for end stations. Additional informative explanations are provided in Annex D.17.
- 39 c) Support the Flow Metering as specified in [Q] 8.6.5.5. Additional informative explanations are
- 40 provided in Annex D.19.

1 5.8 Station Requirements C (SR-C)

2 A station claiming conformance to SR-C shall:

- 3 a) Provide the capabilities for the strict priority algorithm ([Q] 8.6.8 b)).
- 4 b) Support the Credit Based Shaper as specified in [Q] 8.6.8.2. Additional informative explanations are
5 provided in Clause 7.
- 6 c) Support Asynchronous Traffic Shaping as specified in [Q] 8.6.5.2.2, [Q] 8.6.8.5, and [Q] 8.6.11.
7 Additional informative explanations are provided in Clause 6.

8 5.9 Station Requirements D (SR-D)

9 A station claiming conformance to SR-D may:

- 10 a) Support Time Synchronization as stated in Annex D.5.
- 11 b) Support the Enhancements for Scheduled Traffic as specified in [Q] 8.6.8.4 for bridges, and [DC]
12 7.2.4 for end stations. Additional informative explanations are provided in Clause 8.
- 13 c) Support Stream gates per [Q] 8.6.5.4 for bridges, and [DC] 7.2.1.2 for end stations. Additional
14 informative explanations are provided in Annex D.18.2.
- 15 d) Support a maximum gate control list length greater than zero for stream gates as per [Q] 8.6.5.4 for
16 bridges, and [DC] 7.2.1.2 for end stations.

17 5.10 Station Requirements E (SR-E)

18 A station claiming conformance to SR-E may:

- 19 a) Support the preemptable MAC as per item a) of [Q] 6.7.1, and [DC] 7.2.1.2 for end stations.
20 Additional informative explanations are provided in Annex D.25.1.
- 21 b) Support the express MAC as per item b) of [Q] 6.7.1, and [DC] 7.2.1.2 for end stations. Additional
22 informative explanations are provided in Annex D.25.2.

6. Asynchronous Traffic Shaper (ATS)

6.1 Purpose

This clause provides configuration (i.e., via the managed objects specified in the base standards) guidelines for system integrators for the Asynchronous Traffic Shaper (ATS) in order to simplify configuration for in-vehicle networks and avoid errors due to misconfiguration.

This clause provides interpretations of normative features specified in the base standards as listed in Clause 2, and guidance to usage of such features.

Implementations conformant to this standard solely need to conform to the corresponding normative specification in the base standard that specifies a particular feature and is referred according to Clause 5 of this standard. The interpretations described in this clause do not imply requirements and options for claiming conformance to this standard.

Any deviations imposed by this clause from the base standards in terms of additional definitions, limitations, and alterations in this clause beyond the definitions in the base standards do not apply for conforming to the base standards. Implementations solely following this clause instead of the definitions in the base standards cannot claim conformance to the respective features from the base standards. Whenever a conflict between this clause and the normative specification of features in the base standards arise, the normative specification takes precedence over this clause.

6.2 ATS Configuration parameters

ATS is configured through:

- a) Committed Information Rate (ATS-CIR) per ATS Scheduler Instance
- b) Committed Burst Size (ATS-CBS) per ATS Scheduler Instance
- c) Maximum Residence Time (ATS-MRT) per ATS Scheduler Group
- d) Parameters for configuring the ATS Scheduler Group membership of each ATS Scheduler Instance
- e) Additional configuration parameters for mapping received frames to ATS Scheduler Instances (bridges only) and egress TCs

6.3 Implicit policing

ATS has implicit policing capabilities, similar to those of flow meters (D.19). To some extent, flow meter configuration parameter values can be translated to those of ATS scheduler instances and vice versa, as follows:

- a) $MEF-CIR(ATS) = ATS-CIR$
- b) $MEF-CBS(ATS) = ATS-CBS$
- c) $MEF-EIR(ATS) = 0$
- d) $MEF-EBS(ATS) = 0$
- e) $MEF-CF(ATS) = \text{False (0)}$

1 f) MEF-CM(ATS) = color-blind

2 NOTE—The *length(frame)* parameter ([Q] 8.6.11.3.11) used in the ATS ProcessFrame procedure ([Q] 8.6.11.3) includes
3 the media-dependent overhead ([Q] 12.4.2.2), while the flow meter (D.19) only has the MSDU size available for
4 policing. Any resulting deviations are not accounted for in the current standard.

5 In bridges, max. SDU Size filtering is configured for all frames that can be processed by an ATS Scheduler
6 Instance such that frames with a frame size greater than ATS-CBS are discarded before being processed by
7 the Scheduler Instance.

8 Additional Flow Meter policing is not required for Frames to be processed by an ATS Scheduler Instance.

9 NOTE—In bridges, frames in the TC queues can be subject to frame discarding due to frame lifetime limitations ([Q]
10 6.5.6).

11 6.4 ATS Traffic Class Mapping Configuration in Bridges

12 In bridges, the configuration parameters for mapping received frames to ATS Scheduler Instances and from
13 ATS Scheduler Instances to egress TCs are set as follows:

- 14 a) For Frames being processed by an ATS Scheduler Instance, by default no IPV assignment is
15 activated in the Stream Gates.
- 16 b) The *priority* parameter associated with the ingress Frame is therefore mapped to the egress TC
17 Queue in Queuing Frames.
- 18 c) Frames being processed by an ATS Scheduler Instance are queued into the same numerical TC on
19 every hop as they traverse the network.
- 20 d) Frames that arrive on different ports are not processed by the same ATS Scheduler Instance.
- 21 e) Frames that use a different TC anywhere in the network are not processed by the same ATS
22 Scheduler Instance.

23 Considerations for applications in end stations with ATS:

- 24 f) If applications in an end station generate frames, then traffic shaping, e.g., through an ATS
25 Scheduler Instance per application, can provide proper egress behavior at the end station (see also
26 [Q] 47).
- 27 g) In an end station the Applications can be identified as the equivalent to the ingress ports of a bridge.
28 If no Middleware is present and every Application generates Frames just for its own communication
29 needs,
- 30 h) In cases where data from different Applications with different operational cycles is aggregated into
31 Frames by a Middleware (e.g. AUTOSAR's nPDU Feature in [B6] 7.2.2), the trigger conditions in
32 the Middleware can be configured such that proper egress behavior is provided. Combining
33 Middleware triggering and lower layer shaping can create hard to predict egress behavior and is to
34 be avoided.

35

1 7. Credit Based Shaper (CBS)

2 7.1 Purpose

3 This clause provides configuration (i.e., via the managed objects specified in the base standards) guidelines
4 for system integrators for the Credit based Shaper (CBS) in order to simplify configuration for in-vehicle
5 networks and avoid errors due to misconfiguration.

6 This clause provides interpretations of normative features specified in the base standards as listed in
7 Clause 2, and guidance to usage of such features.

8 Implementations conformant to this standard solely need to conform to the corresponding normative
9 specification in the base standard that specifies a particular feature and is referred according to Clause 5 of
10 this standard. The interpretations described in this clause do not imply requirements and options for
11 claiming conformance to this standard.

12 Any deviations imposed by this clause from the base standards in terms of additional definitions, limitations,
13 and alterations in this clause beyond the definitions in the base standards do not apply for conforming to the
14 base standards. Implementations solely following this clause instead of the definitions in the base standards
15 cannot claim conformance to the respective features from the base standards. Whenever a conflict between
16 this clause and the normative specification of features in the base standards arise, the normative
17 specification takes precedence over this clause.

18 7.2 Credit Based Shaper configuration

19 The following parameters are configured for each TC using CBS:

- 20 a) Idle Slope (CBS-IS)
- 21 b) Transmission selection algorithm: CBS

22 Additionally the following information is needed in order to configure a CBS flow:

- 23 c) *portTransmitRate*, the line-rate of the egress port
- 24 d) the media-dependent overhead ([Q] 12.4.2.2) of the egress port
- 25 e) *maxInterferenceTime*, the equivalent in time to [Q] L.1 d) to allow for a more generalized
26 description

27 7.3 Policing Configuration

28 The CBS does not provide implicit Policing. Any Frame to be processed in a CBS TC Queue is subject to a
29 max. SDU Size Filter and a Flow Meter.

30 Suggested Flow Meter policing for CBS Traffic:

- 31 a) $MEF-CIR(CBS) = CBS-IS$
- 32 b) The MEF Committed Burst Size: $MEF-CBS(CBS) = maxBurstSize$
33 where $maxBurstSize = (\text{max. SDU size} + \text{media-dependent overhead})$
34 $+ maxInterferenceTime * portTransmitRate * CBS-IS / (portTransmitRate - CBS-IS)$
35 (derived from [Q] (L-4))
- 36 c) $MEF-EIR(CBS) = 0$

- 1 d) MEF-EBS(CBS) = 0
- 2 e) MEF-CF(CBS) = False (0)
- 3 f) MEF-CM(CBS) = color-blind

4 If different streams are combined to be processed by a single CBS instance and are rate limited by a single
5 flow meter (D.19), then the MEF-CIR becomes the sum of all stream rates (CBS-IS).

6 If the streams go through separate flow meters (D.19) then each stream needs to be configured with its own
7 MEF-CIR, while the CBS uses the sum of all stream rates (CBS-IS).

8 The latency calculations in [BA] 6.6 assume packet size and IPG (Figure C-1) to be included in the
9 bandwidth. The flow meter (D.19) only has the MSDU size available for policing. The resulting deviations
10 are not accounted for in the current standard.

11 7.4 Configuration Rules

12 As can be derived from Annex L of [Q], CBS can create undesired Buffer occupancy and latency in bridges.

13 Traffic that was shaped together in one CBS instance on the previous (upstream) bridge but uses different
14 egress ports on this bridge, is configured with the aggregated bandwidth on this port and on the further
15 downstream path in order to avoid undue shaping delays.

16 Traffic egressing a bridge on the last link to the Listener need not be shaped using CBS.

17 NOTE—This might require more buffer in the end station and cause delays for lower priority traffic, while shaping can
18 cause additional delays for the CBS shaped traffic [Q] Annex L.3.1.3.

1 8. Enhancements for Scheduled Traffic ([Q] 8.6.8.4)

2 8.1 Purpose

3 This clause provides configuration (i.e., via the managed objects specified in the base standards) guidelines
4 for system integrators for the Enhancements for Scheduled Traffic (EST) in order to simplify configuration
5 for in-vehicle networks and avoid errors due to misconfiguration.

6 This clause provides interpretations of normative features specified in the base standards as listed in
7 Clause 2, and guidance to usage of such features.

8 Implementations conformant to this standard solely need to conform to the corresponding normative
9 specification in the base standard that specifies a particular feature and is referred according to Clause 5 of
10 this standard. The interpretations described in this clause do not imply requirements and options for
11 claiming conformance to this standard.

12 Any deviations imposed by this clause from the base standards in terms of additional definitions, limitations,
13 and alterations in this clause beyond the definitions in the base standards do not apply for conforming to the
14 base standards. Implementations solely following this clause instead of the definitions in the base standards
15 cannot claim conformance to the respective features from the base standards. Whenever a conflict between
16 this clause and the normative specification of features in the base standards arise, the normative
17 specification takes precedence over this clause.

18 8.2 Configuration modes

19 This clause (Clause 8) considers particular configuration schemes for EST ([Q] 8.6.8.4), denoted as
20 configuration modes. These modes are the following:

- 21 a) Bus Mode (8.3)
- 22 b) Phased Mode (8.4)
- 23 c) Cyclic Queuing and Forwarding - CQF (8.5)

24 Bus Mode and Phased Mode target minimizing latency. CQF targets reduced jitter at the cost of latency.

25 If EST is used on a port, by default no other shapers are used on TCs whose gates open and close for that
26 port.

27 If EST is used for Scheduled Traffic (Annex C.7.2) only one single TC for Scheduled Traffic is open for
28 transmission at any time. All TCs for non Scheduled Traffic are open during a common time interval in
29 between the Scheduled Traffic TCs openings.

30 If EST is used, the transmission windows are configured so the time synchronization messages can be
31 transmitted at the desired message intervals (within allowed tolerances) in order to achieve the required time
32 synchronization accuracy.

33 NOTE—Computation of optimal schedules for EST throughout the network can, like for most other Time Division
34 Multiple Access (TDMA) systems, be an NP-hard problem [B10].

1 8.3 Bus Mode

2 The intention of Bus Mode is to replicate the behavior one would get from a shared medium being accessed
3 by synchronized Stations on a shared TDMA schedule. Bus Mode opens a communication path from a
4 single Talker to all potential Listeners in the network. The gate opening on all ports between the Talker and
5 the Listeners is intended to be as close as possible to the time at which an application in the Talker generates
6 a frame in order to avoid a waste of bandwidth. The open time needs to be sufficient for the data to traverse
7 all links and bridges between the Talker and the Listeners including store and forward delays. This typically
8 results in sub-optimal bandwidth utilization, especially when different line-rates are mixed.

9 The simplest Bus Mode configuration involves only two intervals:

- 10 a) A short interval
- 11 b) A long interval

12 During the short interval, time critical frames are transmitted from one or more talkers to all relevant
13 listeners. During the long interval, non-time critical frames can be transmitted. The duration of the long
14 interval contributes to the maximum end to end latency of time critical frames. The duration of the short
15 interval, including a configured guard-band ([Q] Q.2), contributes to the maximum end to end latency of
16 non-time critical frames. The topology diameter determines the duration of the short interval. An increasing
17 number of hops results in an increased duration of the short interval. If the exact number of hops is
18 unknown, a conservative maximum may be estimated.

19 8.3.1 Bus Mode Configuration

20 The configuration uses at least 8 GateControlList entries per port.

21 Optimization of a clock tick implementation for bandwidth efficiency includes consideration of link speed
22 (TickGranularity in [Q] 8.6.9.4.16).

23 8.4 Phased Mode

24 In order to allow data to flow from a talker to a listener on a pre-defined path at the lowest possible latency
25 (see protected windows [Q] Q.2), while trying to not block the communication on other paths for longer than
26 needed, each bridge involved is configured with a guard-band and schedule to allow an incoming frame to
27 immediately be transmitted on the destination port.

28 8.4.1 Phased Mode Configuration

29 The configuration uses at least 32 GateControlList entries per port.

30 Optimization of a clock tick implementation for bandwidth efficiency includes consideration of link speed
31 (TickGranularity in [Q] 8.6.9.4.16).

32 8.5 Cyclic Queuing and Forwarding (CQF)

33 This mode is described in [Q] Annex T.

34 NOTE—Bridges need to be able to store frames received in each interval, likely leading to short intervals or large buffer
35 requirements. Upon opening the transmission gate the frames are burst on a strict priority basis. It is possible to use CBS
36 in order to intersperse the frames during transmission, but this does not affect the given (maximum) bounded latency.

1 8.6 EST Latency and Efficiency Considerations

2 One can refer to the Bus Mode and the Phased Mode as minimum latency configurations, while Cyclic
3 Queuing and Forwarding (CQF) is to be viewed as a bounded maximum latency configuration.

4 For Bus Mode and Phased Mode, a frame's worst-case latency caused by EST at a bridge port largely
5 depends on the time the gate for the frame's stream is closed (*operCycleTime* - *gateOpenTime*). This worst-
6 case latency typically occurs only at the first hop or inside the talker. If the incoming frame is already
7 transmitted in sync with network schedule, the latency can be reduced. For these modes, the bandwidth
8 available for a traffic class is related to the ratio of *operCycleTime* over *gateOpenTime*.

9 NOTE—*gateOpenTime* is equal to the total amount of time the gate is open during one cycle.

10 Note that frequent opening and closing of gates is costly irrespective of the mode because of the associated
11 guard-bands. Imprecise synchronization can additionally require over-provisioning of the EST-schedule
12 (*gateOpenTime* in *GateControlList*) and lead to less efficient utilization of the available bandwidth.

13 In a Talker an alignment latency is introduced if applications attempt to transmit data while the gate in the
14 network interface is closed for said traffic. Lowest latency can only be achieved if applications in Talkers
15 and Listeners are synchronized to the EST-schedule of the network, i.e., supply data only at times when
16 transmission is possible and consume data immediately when it is received.

17 8.7 Combining EST with other Shapers

18 8.7.1 General EST considerations

19 The combination of EST with other traffic shapers can introduce additional complexity and lead to difficult
20 to predict impacts on end-to-end latencies. The text in 8.7.1 through 8.7.3 explains potential consequences
21 of combining EST with other shapers such as CBS or ATS.

22 For Bus Mode and Phased Mode it is obviously counter intuitive to apply other shaping mechanisms to
23 frames that are intended to pass through the network as quickly as possible. In order to achieve bounded
24 latency in any of the three modes, it is vital for the transmission queues for any TC to be empty when its
25 transmission gate closes. For CQF this is a fundamental design criterion ([Q] Annex T.1). Otherwise another
26 *operCycleTime* is added to the frame's latency.

27 This standard refers to traffic matching best with the concepts of Bus Mode and Phased Mode as Scheduled
28 Traffic (Annex C.7.2). This implicitly creates a CQF scenario for all other traffic, as it gets queued up while
29 Scheduled Traffic is transmitted and then needs to be transmitted entirely in the window left for all the other
30 queues. Assuming this window is sufficiently long, traffic arriving during said window also needs to be
31 transmitted before its gate closes again to prevent buffer overruns (i.e. Frame loss) and limit latency. In
32 network calculus this is referred to as a stability condition. For more details also see [B11].

33 8.7.2 Combining CBS with EST

34 According to [Q] 8.6.8.2 d) the *idleSlope* for the CBS algorithm is modified if EST is used on the egress port
35 such that no credit is accumulated when the transmission gate is closed and the credit accumulation rate is
36 increased by a factor equal to the inverse of the open duty cycle of the transmission gate. In other words, the
37 CBS algorithm operates only when the transmission gate is open and the entire assigned bandwidth for the
38 EST cycle time can be transmitted during these intervals. The combination of EST with the CBS algorithm
39 could result in a change in the delay of the traffic from the CBS TC due to some intrinsic characteristics of
40 EST, such as the waiting time for transmission gate openings and potential postponed transmissions due to

1 transmission gate closings. Shaped transmission commences as soon as the gate opens, with a proportionally
2 higher shaping rate. Furthermore if lower priority TCs are open concurrently, then they can transmit frames
3 within the shaping gaps, leading to increased latency for the higher priority TCs.

4 **8.7.3 Combining ATS with EST**

5 ATS shaper instances are unaware of the transmission gate state. Assignment of eligibility times continues
6 the same way upon arrival of the frame, independent of the transmission gate state. This leads to the
7 transmission of a burst of frames (depending on relative priority amongst those TCs that are open
8 concurrently) as soon as the transmission gate opens. While this has less impact on the added latency (no
9 further shaping delay) for the stream under consideration, it blocks frames in lower priority TCs from
10 transmission, thereby increasing their latency.

11

Annex A

(normative)

PCS proforma¹

A.1 Introduction

The supplier of an implementation that is claimed to conform to a particular profile defined in this standard shall complete the corresponding Profile Conformance Statement (PCS) proforma.

The tables do not contain an exhaustive list of all requirements that are stated in the referenced standards; for example, if a row in a table asks whether the implementation is conformant to Standard X, and the answer “Yes” is chosen, then it is assumed that it is possible, for that implementation, to fill out the PCS proforma defined in Standard X to show that the implementation is conformant; however, the tables in this standard will only further refine those elements of conformance to Standard X where particular answers are required for the profiles defined here.

The profiles are not intended to be mutually exclusive; it is possible that a given implementation can support more than one of the profiles defined in this standard. If that is the case, then either the PCS for the implementation should be filled out in order to reflect the support of multiple profiles, or a separate PCS should be filled out to reflect each profile supported.

A completed PCS proforma is the PCS for the implementation in question. The PCS is a statement of which capabilities and options of the protocol have been implemented. The PCS can have a number of uses, including use by the following:

- a) Protocol implementer, as a checklist to reduce the risk of failure to conform to the standard through oversight;
- b) Supplier and acquirer—or potential acquirer—of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard PCS proforma;
- c) User—or potential user—of the implementation, as a basis for initially checking the possibility of interworking with another implementation (note that, while interworking can never be guaranteed, failure to interwork can often be predicted from incompatible PCSs);
- d) Protocol tester, as the basis for selecting appropriate tests against which to assess the claim for conformance of the implementation.

A.2 Abbreviations and special symbols

A.2.1 Status symbols

M	mandatory
O	optional
O.n	optional, but support of at least one of the group of options labeled by the same numeral n is required
X	prohibited

¹ Copyright release for PCS proformas: Users of this standard may freely reproduce the PCS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PCS.

- 1 pred:conditional-item symbol, including predicate identification: see A.3.4
2 $\neg$ logical negation, applied to a conditional item's predicate

3 **A.2.2 General abbreviations**

- 4 N/A not applicable
5 PCS Profile Conformance Statement

6 **A.3 Instructions for completing the PCS proforma**

7 **A.3.1 General structure of the PCS proforma**

8 The first part of the PCS proforma, implementation identification and protocol summary, is to be completed
9 as indicated with the information necessary to identify fully both the supplier and the implementation.

10 The main part of the PCS proforma is a fixed-format questionnaire, divided into several subclauses, each
11 containing a number of individual items. Answers to the questionnaire items are to be provided in the
12 rightmost column, either by simply marking an answer to indicate a restricted choice (usually Yes or No) or
13 by entering a value or a set or range of values. (Note that there are some items where two or more choices
14 from a set of possible answers can apply; all relevant choices are to be marked.)

15 Each item is identified by an item reference in the first column. The second column contains the question to
16 be answered; the third column records the status of the item—whether support is mandatory, optional, or
17 conditional; see also A.3.4. The fourth column contains the reference or references to the material that
18 specifies the item in the main body of this standard, and the fifth column provides the space for the answers.

19 A supplier may also provide (or be required to provide) further information, categorized as either Additional
20 Information or Exception Information. When present, each kind of further information is to be provided in a
21 further subclause of items labeled Ai or Xi, respectively, for cross-referencing purposes, where i is any
22 unambiguous identification for the item (e.g., simply a numeral). There are no other restrictions on its format
23 and presentation.

24 A completed PCS proforma, including any Additional Information and Exception Information, is the
25 Protocol Implementation Conformation Statement for the implementation in question.

26 NOTE—NOTE—Where an implementation is capable of being configured in more than one way, a single PCS may be
27 able to describe all such configurations. However, the supplier has the choice of providing more than one PCS, each
28 covering some subset of the implementation's configuration capabilities, in case that makes for easier and clearer
29 presentation of the information.

30 **A.3.2 Additional information**

31 Items of Additional Information allow a supplier to provide further information intended to assist the
32 interpretation of the PCS. It is not intended or expected that a large quantity will be supplied, and a PCS can
33 be considered complete without any such information. Examples might be an outline of the ways in which a
34 (single) implementation can be set up to operate in a variety of environments and configurations, or
35 information about aspects of the implementation that are outside the scope of this standard but that have a
36 bearing on the answers to some items.

37 References to items of Additional Information may be entered next to any answer in the questionnaire and
38 may be included in items of Exception Information.

1 A.3.3 Exception Information

2 It may occasionally happen that a supplier will wish to answer an item with mandatory status (after any
3 conditions have been applied) in a way that conflicts with the indicated requirement. No preprinted answer
4 will be found in the Support column for this item. Instead, the supplier shall write the missing answer into
5 the Support column, together with an Xi reference to an item of Exception Information, and shall provide the
6 appropriate rationale in the Exception item itself.

7 An implementation for which an Exception item is required in this way does not conform to this standard.

8 NOTE—NOTE—A possible reason for the situation described previously is that a defect in this standard has been
9 reported, a correction for which is expected to change the requirement not met by the implementation.

10 A.3.4 Conditional status

11 A.3.4.1 Conditional items

12 The PCS proforma contains a number of conditional items. These are items for which both the applicability
13 of the item itself, and its status if it does apply—mandatory or optional—are dependent on whether certain
14 other items are supported.

15 Where a group of items is subject to the same condition for applicability, a separate preliminary question
16 about the condition appears at the head of the group, with an instruction to skip to a later point in the
17 questionnaire if the “Not Applicable” (N/A) answer is selected. Otherwise, individual conditional items are
18 indicated by a conditional symbol in the Status column.

19 A conditional symbol is of the form “pred: S” where pred is a predicate as described in A.3.4.2, and S is a
20 status symbol, M or O.

21 If the value of the predicate is true (see A.3.4.2), the conditional item is applicable, and its status is indicated
22 by the status symbol following the predicate: The answer column is to be marked in the usual way. If the
23 value of the predicate is false, the “Not Applicable” (N/A) answer is to be marked.

24 A.3.4.2 Predicates

25 A predicate is one of the following:

- 26 a) An item-reference for an item in the PCS proforma: The value of the predicate is true if the item is
27 marked as supported and is false otherwise;
- 28 b) A predicate-name, for a predicate defined as a Boolean expression constructed by combining item-
29 references using the Boolean operator OR: The value of the predicate is true if one or more of the
30 items is marked as supported;
- 31 c) The logical negation symbol “¬” prefixed to an item-reference or predicate-name: The value of the
32 predicate is true if the value of the predicate formed by omitting the “¬” symbol is false, and vice
33 versa.

34 Each item whose reference is used in a predicate or predicate definition, or in a preliminary question for
35 grouped conditional items, is indicated by an asterisk in the Item column.

1 A.4 Bridge PCS

Table A-1—Bridge PCS

Item	Feature	Status	References	Support
C-VLAN	Does the bridge support C-VLAN tags?	M	5.4.1:a)	Yes []
B-SR-A	Does the bridge support SR-A?	M	5.4.1:b)	Yes []
B-SR-B	Does the bridge support SR-B?	M	5.4.1:c)	Yes []
B-SR-C	Does the bridge support SR-C?	M	5.4.1:d)	Yes []
PCS—1	Does the bridge support the Learning Process?	M	5.4.1:e)	Yes []
PCS—2	Does the bridge support the maximum SDU Size?	M	5.4.1:f)	Yes []
PCS—3	Does the bridge support Ingress Filtering?	M	5.4.1:g)	Yes []
PCS—4	Does the bridge support Egress Filtering?	M	5.4.1:h)	Yes []
PCS—5	Does the bridge support Frame Filtering?	M	5.4.1:i)	Yes []
PCS—6	Does the bridge support an FDB with at least 1024 entries?	M	5.4.1:j)	Yes []
PCS—7	Does the bridge support static IPV configuration?	M	5.4.1:k)	Yes []
PCS—8	Are the bridge's Stream Gates open by default?	M	5.4.1:l)	Yes []
B-TimeSync	Does the bridge support Time Synchronization? If Yes, provide details on the supported Time Synchronization solution: 	SR-D- TimeSync:M	5.4.2:a)	Yes [] No []
B-SR-D	Does the bridge support SR-D?	O	5.4.2:b)	Yes [] No []
B-SR-E	Does the bridge support SR-E?	O	5.4.2:c)	Yes [] No []
B-MACsec	Does the bridge support MACsec?	O	5.4.2:d)	Yes [] No []

1 A.5 End Station PCS

Table A-2—End Station PCS

Item	Feature	Status	References	Support
E-SR-B	Does the end station support SR-B?	M	5.5.1:a)	Yes []
PCS—9	Does the end station support Out-facing Ingress Stream Identification Function(s)?	M	5.5.1:b)	Yes []
PCS—10	Does the end station support at least 64 stream identification functions? If Yes, provide the number of stream identification functions: _____	M	5.5.1:c)	Yes []
PCS—11	Does the end station support Stream Filter assignment?	M	5.5.1:d)	Yes []
PCS—12	Does the end station support max. SDU Size Filtering?	M	5.5.1:e)	Yes []
PCS—13	Does the end station support the ATS Talker transmission behavior on one TC.	M	5.5.2:a)	Yes []
PCS—14	Does the end station support the CBS algorithm on the two numerically highest value TCs.	M	5.5.2:b)	Yes []
PCS—15	Does the end station support Queuing frames?	M	5.5.2:c)	Yes []
PCS—16	Does the end station support 8 traffic classes?	M	5.5.2:d)	Yes []
PCS—17	Does the end station support Transmission selection?	M	5.5.2:e)	Yes []
E-TimeSync	Does the end station support Time Synchronization? If Yes, provide details on the supported Time Synchronization solution: _____	SR-D-TimeSync:M	5.5.3:a)	Yes [] No []
E-SR-D	Does the end station support SR-D?	O	5.5.3:b)	Yes [] No []
E-SR-E	Does the end station support SR-E?	O	5.5.3:c)	Yes [] No []
PCS—18	Does the end station support MACsec?	O	5.5.3:d)	Yes [] No []
PCS—19	Does the end station support Stream Gating on ingress?	O	5.5.3:e)	Yes [] No []
PCS—20	Does the end station support Flow metering on ingress?	O	5.5.3:f)	Yes [] No []

1 A.6 Station Requirements A (SR-A) PCS

Table A-3—SR-A PCS

Item	Feature	Status	References	Support
PCS—21	Does the station's port support port-based VLAN Classification?	B-SR-A:M	5.6:a)	Yes [] No []
PCS—22	Does the station's port support Priority Code Point Decoding?	B-SR-A:M	5.6:b)	Yes [] No []
PCS—23	Does the station's port support Priority Regeneration?	B-SR-A:M	5.6:c)	Yes [] No []
PCS—24	Does the station's port support Ingress Stream Identification Function(s)?	B-SR-A:M	5.6:d)	Yes [] No []
PCS—25	Does the bridge support a sufficient number of Stream Identification Function(s)? If Yes, provide the number of ports: _____ and the number of Stream Identification Function(s): _____	B-SR-A:M	5.6:e)	Yes [] No []
PCS—26	Does the station's port support Active topology enforcement?	B-SR-A:M	5.6:f)	Yes [] No []
PCS—27	Does the station's port support Ingress Filtering?	B-SR-A:M	5.6:g)	Yes [] No []
PCS—28	Does the station's port support Egress Filtering	B-SR-A:M	5.6:h)	Yes [] No []
PCS—29	Does the station's port support Queuing Frames?	B-SR-A:M	5.6:i)	Yes [] No []
PCS—30	Does the station's port support 8 traffic classes?	B-SR-A:M	5.6:j)	Yes [] No []
PCS—31	Does the station's port support Transmission Selection?	B-SR-A:M	5.6:k)	Yes [] No []
PCS—32	Does the station support the ATS transmission selection algorithm on one TC.	B-SR-A:M	5.6:l)	Yes [] No []
PCS—33	Does the station support the CBS algorithm on the two numerically highest value TCs.	B-SR-A:M	5.6:m)	Yes [] No []
PCS—34	Does the station's port support VID Translation and Egress VID Translation?	B-SR-A:M	5.6:n)	Yes [] No []

1 A.7 Station Requirements B (SR-B) PCS

Table A-4—SR-B PCS

Item	Feature	Status	References	Support
PCS—35	Does the station's port support the Acceptable Frame Types Filter?	B-SR-B:M E-SR-B:M	5.7:a)	Yes [] No []
PCS—36	Does the station's port support the Maximum SDU Size Filtering?	B-SR-B:M E-SR-B:M	5.7:b)	Yes [] No []
PCS—37	Does the station's port support Flow Metering?	B-SR-B:M E-SR-B:M	5.7:c)	Yes [] No []

2 A.8 Station Requirements C (SR-C) PCS

Table A-5—SR-C PCS

Item	Feature	Status	References	Support
SR-C-SP	Does the station's port provide the capabilities for the strict priority algorithm?	B-SR-C:M	5.8:a)	Yes [] No []
SR-C-CBS	Does the station's port provide the capabilities for the Credit Based Shaper?	B-SR-C:M	5.8:b)	Yes [] No []
SR-C-ATS	Does the station's port provide the capabilities for the Asynchronous Traffic Shaper	B-SR-C:M	5.8:c)	Yes [] No []

3 A.9 Station Requirements D (SR-D) PCS

Table A-6—SR-D PCS

Item	Feature	Status	References	Support
SR-D-TimeSync	Does the station's port have Time Synchronization activated?	B-SR-D:M E-SR-D:M	5.9:a)	Yes [] No []
PCS—38	Does the station's port provide the capabilities for the Enhancements for Scheduled Traffic?	B-SR-D:O E-SR-D:M	5.9:b)	Yes [] No []
PCS—39	Does the station's port support Stream gates?	B-SR-D:O E-SR-D:M	5.9:c)	Yes [] No []
PCS—40	Does the station's port support a maximum gate control list length greater than zero?	B-SR-D:O E-SR-D:M	5.9:d)	Yes [] No []

1 **A.10 Station Requirements E (SR-E) PCS**

Table A-7—SR-E PCS

Item	Feature	Status	References	Support
SR-D-pMAC	Does the System support the preemptable MAC	B-SR-E:M E-SR-E:M	5.10:a)	Yes [] No []
SR-D-eMAC	Does the System support the express MAC	B-SR-E:M E-SR-E:M	5.10:b)	Yes [] No []

2

1 Annex B

2 (informative)

3 Intentional Limitations

4 B.1 Shaper Interactions

5 In [B11] a line of reasoning is given why combining the ATS (Clause 6) or CBS (Clause 7) mechanisms
6 with EST (Clause 8) may lead to a more complicated configuration in certain network topologies. While [Q]
7 allows such combinations (8.7), this standard does not give guidance on how such configurations affect
8 network behavior.

9 B.2 MACsec [AE]

10 This standard does not give guidance on how to use MACsec [AE], but considers it to be just another
11 protocol to be transported across the network.

12 B.3 Frame Preemption Interactions

13 While [Q] certainly allows combinations of Frame Preemption (D.25) with for example EST (Clause 8), this
14 standard does not give guidance on how such configurations affect network behavior.

15

1 Annex C

2 (informative)

3 Terminology

4 C.1 Automotive Terminology

5 This standard uses terminology that is consistent with automotive uses, without contradicting established
6 IEEE 802 terminology.

7 C.2 Automotive Stations ([802])

8 The IEEE 802.1 family of standards define features for relays, bridges [Q] and end stations ([802]), which
9 are part of the profile definitions in this standard.

10 C.2.1 Automotive Bridge

11 The combination of a Bridge Management Entity [Q], a MAC relay entity and at least two bridge ports is
12 referred to as a bridge ([Q]).

13 The Bridge Management Entity [Q] can act as an end station for certain protocols related to infrastructure
14 services, like e.g. time synchronization, service discovery, or diagnostics.

15 C.2.2 Automotive End Station

16 An end station is the source or the destination of the MAC Client Data ([802.3]) in a frame.

17 C.3 Automotive Electronic Control Unit (ECU)

18 In this standard an ECU is any encased electronic device inside a vehicle. It is linked to other ECUs through
19 power and communication connections.

20 C.4 Communication Aggregates

21 A MAC frame is defined by [802.3] as the core part of a packet (see also C.5). Multiple Packets sharing a
22 common source and destination, but traveling along different paths in a network topology, are considered to
23 belong to Member Streams. A unidirectional Flow of Packets from one source to one or more destinations is
24 referred to as a Stream, as per IEEE Std 802.1Q ([Q]). All Packets that are part of a specific functional
25 communication relation between two or more End Stations are considered to be part of one Flow. Any
26 packet sharing a certain characteristic (ingress port, or VLAN-ID, or destination MAC address, or ...) at a
27 specific point in or along the network topology can be referred to as Traffic.

1 C.5 Frame Content as Received

2 This standard uses the definitions of packet and MAC frame as defined in [802.3] Figure 3-1. The media-
3 dependent overhead of [Q] 12.4.2.2 can be described as the packet size plus the IPG (Figure C-1) minus the
4 frame size. The MSDU for this standard (Figure C-1) is the *mac_service_data_unit* as passed from/to the
5 Enhanced Internal Sublayer Service (EISS) of [Q] 6.8.1.

6 A frame in the context of this standard contains,

- 7 a) One MAC Source Address,
- 8 b) one MAC Destination Address,
- 9 c) an optional C-VLAN Tag (see below),
- 10 d) and one (last) EtherType.

11 The C-VLAN Tag ([Q] 9) is composed of:

- 12 e) A (first/outer EtherType) Tag Protocol Identifier (TPID) of 0x8100
- 13 f) a Priority Code Point (PCP)
- 14 g) a Drop Eligible Indicator bit (DEI)
- 15 h) a VLAN Identifier (VID)

16 C.5.1 Packet and Frame size

17 The bandwidth and latency characteristics of an Ethernet network are sensitive to the packet ([802.3] Figure
18 3-1) sizes that are used in the network. Specifically when analyzing the queuing delay (e.g. [Q] Annex
19 L.3.1.1) of a specific frame X, another frame Y of any priority could have been selected for transmission an
20 arbitrarily small time before frame X became eligible for transmission. The actual packet length of frame Y
21 is dependent on the exact header structure and physical layer overhead (see Figure C-1). All relevant
22 shaping mechanisms (ATS (Clause 6), CBS (Clause 7) and EST (Clause 8)) specifications refer to lengths as
23 including (media-dependent) overhead ([Q] 12.4.2.2). For ATS it is explicitly stated in [Q] 8.6.11.3.11.

1 8.6.8.4, for CBS this can best be seen in [Q] L.2 item b), and for EST in [Q] 8.6.8.4 NOTE 1.

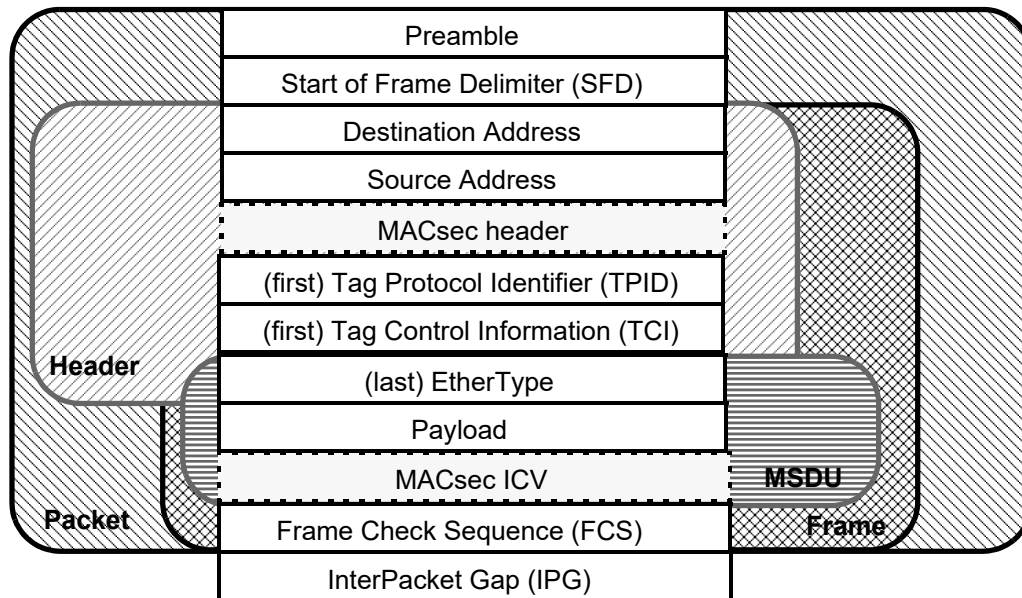


Figure C-1—Example of Packet and Frame formats

2 C.5.2 Untagged Frame ([Q] 3.286)

3 As this standard considers only C-VLAN tags ([Q] 9.5 a)), an Untagged Frame in the context of this
 4 standard carries a value other than 0x8100 in the outermost (first) EtherType (Figure C-1), when processed
 5 in the EISS ([Q] 6.8).

6 C.5.3 VLAN Tagged Frame ([Q] 9.5)

7 As this standard considers only C-VLAN tags ([Q] 9.5 a)), a (VLAN) Tagged Frame ([Q] 3.267) in the
 8 context of this standard carries a TPID of 0x8100 ([Q] Table 9-1) in the outermost (first) EtherType
 9 (Figure C-1) and a non-Zero VID field, when processed in the EISS ([Q] 6.8).

10 C.5.4 Priority-Tagged Frame ([Q] 6.9)

11 As this standard considers only C-VLAN tags ([Q] 9.5 a)), a Priority-Tagged Frame in the context of this
 12 standard carries a TPID of 0x8100 ([Q] Table 9-1) in the outermost (first) EtherType (Figure C-1) and a VID field of all
 13 Zero, when processed in the EISS ([Q] 6.8).

14 C.5.5 MACsec [AE]

15 If MACsec is applied, the SecTAG ([AE] 9.3) is placed between the Source Address and the C-TAG ([Q] 9.5
 16 a)) of the frame (Figure C-1), as depicted in [AE] Figure 6-2.

1 C.6 Bandwidth

2 Bandwidth (B) is in general defined as the quotient of amount of Data (D) over an interval of Time (T):

3 $B = D/T$.

4 Any Ethernet communication link operates at a certain Line Rate (R). While the link is transmitting data, the
5 instantaneous Bandwidth is always equal to the Line Rate. Only if the Time interval (T) includes a period
6 where no data is transmitted on the link, can the Bandwidth of Traffic on that link drop below the Line Rate.
7 It is therefore not advisable to give Bandwidth information without an indication of the period of Time over
8 which it is measured.

9 One can also define Bandwidth as a number of Frames (F) of equal length (L) over an interval of Time (T):

10 $B = F \cdot L / T$.

11 The following factors influence frame size (L):

- 12 a) Header information at all ISO/OSI Layers.
- 13 b) Serialization of transported data.
- 14 c) Safety and Security overhead.

15 The following factors influence the number of Frames (F) over a period of Time (T):

- 16 d) Actual data available for transmission.
- 17 e) Maximum allowed SDU size (causing segmentation or fragmentation).
- 18 f) Retransmissions of lost Frames in reliable Flows (e.g. TCP [B12]).

19 C.7 Network Traffic Classification

20 C.7.1 Automotive Network Traffic Patterns

21 The number of frames on the IVN is dominated by cyclic messages to monitor or control (safety critical)
22 functionality. Their periodicity varies between about 1ms and 500ms, i.e. less than 1000 frames per second
23 per stream. Due to the mostly low to medium frame size (Annex C.5), the average Bandwidth (Annex C.6)
24 demand is low, but there is a large number of such streams, which are critical to the operation of the vehicle.

25 High resolution sensors (cameras, LIDARs, RADARs) can generate (1.5kB) frames (Annex C.5) at a rate of
26 10s of thousands per second. This traffic dominates the Bandwidth (Annex C.6) in the IVN.

27 Acoustic sensors for active noise cancellation can generate a small frame about every 20μs. These can be
28 considered to have the highest latency constraint within the IVN, but at very low Bandwidth.

29 FlexRay [B9] enables a request-response exchange (2 messages) to happen within less than 100μs, i.e.
30 within a single FlexRay cycle.

31 The so called “Best Effort” (BE) Traffic in the vehicle consists of distribution of SotA data, web access by
32 the occupants, map downloads, and similar not time triggered Frames. This is the most bursty traffic in the
33 IVN. In contrast to an IT or home network, this traffic does however have a need to be guarded against
34 losses (C.10). Since a high percentage of traffic in the IVN is dominated by some sort of time triggered
35 events, the Bandwidth available for BE Traffic is not as temporarily varying as in an office or home network

1 installation. Retransmissions due to frame loss present an unpredictable source of bandwidth demand, which
2 in turn increases the risk of frame loss. This kind of feedback loop is best avoided.

3 **C.7.2 Scheduled Traffic**

4 The application independent term “Scheduled Traffic” is to denote streams, where transmission, reception,
5 and response are tightly coupled in the time domain. While no absolute numbers are given, the latency and
6 periodicity can be assumed to be below a few 100µs.

7 **C.8 Scheduler vs. Shaper**

8 This standard defines the terms as follows:

- 9 a) Scheduling: Is the action of assigning Resources
- 10 b) Traffic Shaping: Is a Bandwidth management technique
- 11 c) Policing: Is the rule for bandwidth based discarding of frames

12 If the Resource assigned by a Scheduler is (access to) Bandwidth, then a Scheduler implicitly performs
13 Traffic Shaping. E.g.: The Enhancements for Scheduled Traffic (EST of Clause 8) is based on Scheduling
14 access to a transmission port based on a timetable.

15 Strict Priority Queuing (SPQ) assigns access to a transmission port based on priority. But SPQ does not
16 manage the bandwidth, as the highest priority can get 100% access to the transmission port.

17 An ingress Rate Limiter discards frames (policies them) in case they arrive too frequently. Less bandwidth is
18 required for the left over egress traffic, but one would usually not call this shaping.

19 **C.9 FRER Functionality**

20 Frame Replication and Elimination for Reliability (FRER) as well as Stream Identification are defined in
21 IEEE Std 802.1CB ([CB]).

22 In the context of this standard the FRER Functionality is referring to all those functions defined in [CB],
23 except the Stream Identification functions ([CB] 6).

24 In particular FRER Functionality entails:

- 25 a) Sequencing function ([CB] 7.4)
- 26 b) Individual recovery function ([CB] 7.5)
- 27 c) Sequence encode/decode function ([CB] 7.6)
- 28 d) Stream splitting function ([CB] 7.7)
- 29 e) Redundancy Tag ([CB] 7.8)
- 30 f) Sequence information ([CB] 7.9 and [CB] 7.10)

1 **C.10 Frame Loss**

2 A frame can be lost if the Relay is unable to process it:

- 3 a) Due to an invalid FCS on ingress. ([Q] 6.5.2 a))
- 4 b) If the SDU size ([802.3] 3.2.7) of the frame exceeds the maximum supported SDU size of the PHY/
5 MAC on egress. ([Q] 6.5.2 b)3))
- 6 c) If the SDU size ([802.3] 3.2.7) of the frame exceeds the maximum supported SDU size of the
7 Relay's IEEE 802.1 Features on ingress.
- 8 d) If the SDU size ([802.3] 3.2.7) of the frame is below the minimum supported SDU size of the PHY/
9 MAC on ingress. ([802.3] 4.2.3.3)

10

Annex D

(informative)

IEEE 802.1 Features

D.1 Purpose

This annex provides interpretations of normative features specified in the base standards as listed in Clause 2, and guidance to usage of such features.

Implementations conformant to this standard solely need to conform to the corresponding normative specification in the base standard that specifies a particular feature and is referred according to Clause 5 of this standard. The interpretations described in this annex do not imply conformance requirements.

Any deviations imposed by this annex from the base standards in terms of additional definitions, limitations, and alterations in this annex beyond the definitions in the base standards do not apply for conforming to the base standards. Implementations solely following this annex instead of the definitions in the base standards cannot claim conformance to the respective features from the base standards. Whenever a conflict between this annex and the normative specification of features in the base standards arise, the normative specification takes precedence over this annex.

D.2 Bridge Processing

The following list gives a summary of processing operations which can be executed by a bridge.

- a) Default Priority Assignment ([AC] 13.1)
- b) MACsec processing ([AE] 11.4)
- c) Support of the EISS ([Q] 6.9)
- d) Frame Type Acceptance filter ([Q] 6.9 c))
- e) Out-facing Ingress Stream Identification Function(s) ([CB] 9.1.1.5)
- f) Port-based VLAN Classification ([Q] 6.9 d))
- g) Priority Code Point Decoding ([Q] 6.9.3)
- h) Priority Regeneration ([Q] 6.9.4)
- i) Active topology enforcement ([Q] 8.6.1)
- j) Ingress filtering ([Q] 8.6.2)
- k) Frame filtering ([Q] 8.6.3)
- l) Egress filtering ([Q] 8.6.4)
- m) Stream filter assignment ([Q] 8.6.5.3 b) and c))
- n) Maximum SDU Size Filtering ([Q] 8.6.5.3.1)
- o) Stream Gating ([Q] 8.6.5.4)
- p) Flow metering ([Q] 8.6.5.5)
- q) ATS Eligibility Time Assignment ([Q] 8.6.5.6)
- r) In-facing Egress Stream Identification Function(s) ([CB] 9.1.1.4)
- s) FRER Functionality (C.9)
- t) Out-facing Egress Stream Identification Function(s) ([CB] 9.1.1.3)

- 1 u) Queuing frames ([Q] 8.6.6)
- 2 v) Queue Management (D.24)
- 3 w) Transmission selection ([Q] 8.6.8)
- 4 x) Priority Code Point Encoding ([Q] 6.9.3)
- 5 y) Egress VID translation ([Q] 6.9 g))
- 6 z) Support of the ISS ([Q] 6.7.1)

7 NOTE—Potentially not all listed feature are required, depending on the Station Requirements (Clause 5) applied.

8 **D.3 End Station Receive Path Processing Order (ingress)**

9 Frames are processed in the receive path of an end station by the listed features, if they are activated.

- 10 a) MACsec processing ([AE] 11.2)
- 11 b) Frame Type Acceptance filter ([Q] 6.9 c))
- 12 c) Out-facing Ingress Stream Identification Function(s) ([CB] 9.1.1.5)
- 13 d) Stream filter assignment ([Q] 8.6.5.3 b)
- 14 e) Maximum SDU Size Filtering ([Q] 8.6.5.3.1)
- 15 f) Stream Gating ([Q] 8.6.5.4)
- 16 g) Flow metering ([Q] 8.6.5.5)

17 NOTE—Potentially not all listed feature are required, depending on the Station Requirements (Clause 5) applied.

18 **D.4 End Station Transmission Path Processing Order (egress)**

19 Frames are processed in the transmission path of an end station by the listed features, if they are activated.

- 20 a) Maximum SDU Size Filtering ([Q] 8.6.5.3.1)
- 21 b) Shaping as per 5.5.2 item a)
- 22 c) Queuing frames ([Q] 8.6.6)
- 23 d) Transmission selection ([Q] 8.6.8)
- 24 e) MACsec processing ([AE] 11.4)

25 NOTE—Potentially not all listed feature are required, depending on the Station Requirements (Clause 5) applied.

26 **D.5 Time Synchronization**

27 Time synchronization is an essential component of in-vehicle use-cases of this standard; however, this
 28 standard allows different mechanisms to be used for time synchronization, as different mechanism have
 29 already been deployed in the field. Some examples of time synchronization specifications that can be used in
 30 applications requiring time synchronization are as follows:

- 31 a) IEEE Std 802.1AS-2020 [B1]
- 32 b) IEEE Std 802.1AS-2011 [B2]

- 1 c) AVnu Automotive profile [B13]
- 2 d) AUTOSAR Time Synchronization Protocol [B7]
- 3 e) IEEE Std 1588-2019 ([B8] and its profiles: <https://sagroups.ieee.org/1588/ptp-profiles/>)

4 NOTE—It is up to the system integrator to select time synchronization solutions that meet their interoperability and
5 application requirements.

6 D.6 MACsec processing ([AE])

7 This standard does not address the details of a MACsec implementation. If an integrator chooses to
8 implement MACsec in (parts of) the network, they at least need to consider MACsec's impact on for
9 example Frame Preemption (5.10), EST gate schedule (5.9), Shapers (5.8), and policing (5.7). As this list is
10 not claiming completeness, the implementer is advised to consider further implications.

11 Annex C.5.5 describes the placement of the SecTAG in the frame's header.

12 On a C-VLAN aware port, on ingress the SecTAG is processed before the C-VLAN Tag, if MACsec is
13 applied [AE].

14 D.7 Acceptable Frame Types ([Q] 6.9 c))

15 The Acceptable Frame Types parameter can take the following values:

- 16 a) Admit Only VLAN-tagged frames (C.5.3)
- 17 b) Admit Only Untagged and Priority-tagged frames (C.5.2, and C.5.4, respectively)
- 18 c) Admit All frames

19 Frames not matching the Acceptable Frame Types parameter are discarded silently.

20 D.8 Port-based VLAN Classification ([Q] 6.9.1 f))

21 If Port-based VLAN classification is activated, and Port-and-Protocol-based VLAN classification is not
22 activated, and:

- 23 a) the Frame is Untagged (C.5.2), or
- 24 b) Priority-tagged (C.5.4), or
- 25 c) the VID translation table is activated and the translation already set the *vlan_identifier* parameter to
26 Zero,

27 the *vlan_identifier* parameter is set/changed to the Port-VID (PVID) value for the port.

28 NOTE 1—Port-and-Protocol-based VLAN classification is not required by this standard, but could still be active in the
29 device.

30 NOTE 2—The VID translation table is not required by this standard, but could still be active in the device.

31 D.9 Priority Code Point Decoding ([Q] 6.9.3)

32 The *drop_eligible* parameter is set to the received DEI Field's value (1 bit).

33 The *priority* parameter is set to the received PCP Field's value (3 bit).

1 D.10 Priority Regeneration ([Q] 6.9.4)

2 If the Priority Regeneration is activated, the *priority* is set/changed according to the Priority Regeneration
3 configuration.

4 D.11 Ingress Stream Identification Function(s) ([CB] 6.2)

5 D.11.1 Stream Identification Function(s) for Bridges

6 Only passive out-facing stream identification functions are configurable, with the Identification Mask
7 Length as stated in D.11.3.

8 NOTE 1—Please refer to the informative Annex F on Stream Identification for a rationale why only out-facing ingress
9 stream identification functions are supported.

10 The Stream Identification functions operate on the *M_UNITDATA.indication* parameters of the Internal
11 Sublayer Service (ISS) ([AC] 12.2).

12 All frames passing through the stream identification functions in the UP direction are assigned a
13 *stream_handle*. A *stream_handle* of value *Null* (a negative denotes the *Null* value) indicates no matching
14 stream identification function was found.

15 Bridges with the number of ports as given in Table D-1 support the listed number of identification functions.

Table D-1—Identification profile for Bridges

Number of ports (<i>n</i>)	Minimum number of Identification functions
less than 5 ($2 < n < 5$)	128
more than 4, but less than 10 ($4 < n < 10$)	192
more than 9 ($n > 9$)	256

16 D.11.2 Stream Identification Function(s) for End Stations

17 Only passive out-facing stream identification functions are configurable, with the Identification Mask
18 Length as stated in D.11.3.

19 A minimum number of 64 Identification functions is supported on an ingress port.

20 D.11.3 Identification Mask Length ([CBdb])

21 All filter masks are long enough to cover:

- 22 — MAC addressing
- 23 — one VLAN tag
- 24 — the addresses, DSCP, and next header field of an IPv6 and protocol field of an IPv4 header
- 25 — the port information of TCP and UDP

26 NOTE 1—IPv6 extension headers and IPv4 options are out of scope, as they are not covered by [CB].

1 **D.12 Active topology enforcement ([Q] 8.6.1)**

2 If ingress filtering ([Q] 8.6.2) did not cause the received frame to be discarded, the source address and VID
3 are submitted to the Learning Process (D.14). The Learning Process can be configured by management,
4 under which conditions learning is enabled or disabled is not further specified here.

5 A loop-free network topology is to be provided through configuration without requiring the use of the Rapid
6 Spanning Tree Algorithm and Protocol.

7 The forwarding of any frames to a port can be disabled by management. The conditions under which
8 management allows or disables the forwarding are not further specified here.

9 **D.13 Ingress Filtering ([Q] 8.6.2)**

10 Frames received on a port that is not in the member set ([Q] 8.8.10) associated with the Frame's associated
11 *vlan_identifier* parameter are discarded.

12 **D.14 The Learning Process ([Q] 8.7)**

13 The Learning Process:

- 14 a) Supports shared C-VLAN learning of source MAC addresses ([Q] Annex F).
- 15 b) Supports independent C-VLAN learning of source MAC addresses ([Q] Annex F).

16 **D.15 Frame Filtering ([Q] 8.6.3)**

17 **D.15.1 Introduction to Frame Filtering**

18 The set of potential transmission ports ([Q] 8.6.1) for each received Frame is reduced on the basis of:

- 19 a) The *destination_address* parameter ([Q] 8.6.3 a))
- 20 b) The *vlan_identifier* parameter ([Q] 8.6.3 b))
- 21 c) The Filtering Database (FDB) entries ([Q] 8.6.3 d))
- 22 d) The default Group Filtering behavior ([Q] 8.6.3 e))

23 **D.15.2 Reserved Addresses (01-80-C2-...)**

24 The Reserved Addresses of [Q] Table 8-1 and [Q] Table 8-2 are supported.

25 NOTE—Frames addressed to Reserved Addresses can carry a C-VLAN Tag on wire and have an associated
26 *vlan_identifier* parameter.

27 **D.15.3 Bridge Protocol Data Units (BPDUs) ([Q] 14)**

28 BPDUs are forwarded to a defined management port. The further handling of BPDUs is currently not
29 mandated by this standard.

1 **D.16 Stream Filter ([Q] 8.6.5.3)**

2 A Stream Filter is identified by

- 3 a) a *stream_handle* specification ([Q] 8.6.5.3 b)) AND
- 4 b) a *priority* specification ([Q] 8.6.5.3 c)).

5 Frames assigned to the same Stream Filter are processed together in the same SF-Instances of Max. SDU
6 Size Filtering (D.17.3), Stream Gating (D.18), Flow Metering (D.19), and ATS eligibility time assignment
7 ([Q] 8.6.5.6).

8 More than one stream filter instance can be configured to use a specific Stream Gating, Flow Metering, and
9 ATS eligibility time assignment instance.

10 A Frame with multiple egress ports passes only through a single Stream Filter.

11 A certain instance can be configured for one or more Stream Filters, meaning Frames with different
12 associated *stream_handle* parameters might be processed by the same SF-Instance.

13 If the Frames are intended to go through a CBS, a Stream Filter configuration is suggested in Clause 7.

14 If the Frames are intended to go through an ATS, a Stream Filter configuration is suggested in Clause 6.

15 **D.17 Maximum SDU Size Filtering**

16 **D.17.1 Introduction to SDU Size Filtering**

17 The guarantee of an upper bound for the SDU size is required for the calculation of worst-case delays. The
18 text in D.17.1 through D.17.4 explains how to configure the necessary Maximum SDU size Filters.

19 **D.17.2 Service Data Unit (SDU) Size**

20 The SDU size considered here is specifically the MSDU size (Figure C-1), as described in Annex C.5.

21 All IEEE 802.1 features support at least a maximum MSDU size of 1982 octets, to allow for envelope
22 frames ([802.3] 1.4.3.10).

23 **D.17.3 Per Stream Filter Maximum SDU Size ([Q] 8.6.5.3.1)**

24 All stream filters have a Max SDU Size filter configured to the max of 1982 octets (Annex D.17.2) or a user
25 selected value.

26 **D.17.4 Per Traffic Class Maximum SDU Size ([Q] 8.6.8.4)**

27 A per TC max. SDU size filter can be configured and activated, even if no gate schedule is configured, i.e.
28 even if all transmission gates are always open.

1 **D.18 Stream Gates ([Q] 8.6.5.4)**

2 **D.18.1 Static IPV configuration**

3 A static *IPV* assignment per Stream Gate can be configured.

4 The allowed range for the *IPV* parameter is not specified explicitly in [Q]. In practice, stations can use a
5 number of methods to assign a frame an *IPV*, and thus to an output queue. This standard does not restrict
6 such behavior (see also Annex D.20).

7 **D.18.2 Time dependent gate states**

8 Based on a separate schedule from the transmission gates, the Stream Gates are assuming either the *OPEN*
9 or the *CLOSED* state. If a frame arrives while its associated stream gate is closed, it is discarded. A counter
10 for discarded and passed frames can be available.

11 According to [Q] 8.6.10.4, the default state of a stream gate is *OPEN*. For any scenario where stream gating
12 is not used, the stream gates can be left in their default state.

13 **D.19 Flow Metering ([Q] 8.6.5.5)**

14 All Stream Filters have a Flow Meter configured.

15 The MEF algorithm referenced in [Q] 8.6.5.5 is supported with:

- 16 a) A configurable Committed Information Rate: $\text{MEF-CIR} > 0$
- 17 b) A configurable Committed Burst Size: $\text{MEF-CBS} > 0$
- 18 c) A Excess Information Rate: $\text{MEF-EIR} = 0$
- 19 d) A Excess Burst Size: $\text{MEF-EBS} = 0$
- 20 e) A Coupling Flag: $\text{MEF-CF} = \text{False} (0)$
- 21 f) A Color Mode Flag: $\text{MEF-CM} = \text{color-blind}$

22 Items a) through e) of the above list represent a Single Rate Two Color Meter. Item f) makes it color-blind on
23 ingress, which is consistent with the egress behavior.

24 Frames are either:

- 25 g) Permitted to pass (green) OR
- 26 h) Dropped (red)

27 At this processing stage only the MSDU size (Annex C.5) is known, i.e. neither the media-dependent
28 overhead ([Q] 12.4.2.2) at ingress nor at egress are known. It is therefore not possible to easily deduce the
29 actual bandwidth used by the packet on wire.

30 **D.20 Queuing Frames ([Q] 8.6.6)**

31 Queuing frames is performed per egress port.

32 The default priority to traffic class (TC) mapping is performed according to [Q] Table 34-1.

1 Each frame is mapped to a traffic class using the Traffic Class Table for the respective egress port. The
2 parameters used for this mapping are determined as follows:

- 3 a) If stream gates ([Q] 8.6.5.4) are not supported, the frame's associated *priority* parameter is used.
- 4 b) If stream gates are supported and the *IPV* parameter assigned to the frame is *Null*, the frame's
5 associated *priority* parameter is used.
- 6 c) If stream gates are supported and the *IPV* parameter assigned to the frame is Non-*Null*, the *IPV*
7 parameter is used (see also D.18.1).

8 NOTE—A negative value for the *IPV* denotes the *Null* value.

9 **D.21 Transmission Selection**

10 **D.21.1 Per TC-Queue Transmission Selection ([Q] 8.6.8 a))**

11 The operation of the transmission selection algorithm activated for a given TC Queue determines if there is
12 a frame available for transmission or not.

- 13 a) At least one TC Queue on any port support the ATS Transmission Selection algorithm of [Q] 8.6.8.5.
- 14 b) At least the two numerically highest value Traffic Class Queues (highest priority) on any port
15 support the Credit Based Shaper transmission selection of [Q] 8.6.8.2.

16 **D.21.2 Port Transmission Selection ([Q] 8.6.8 b))**

17 For each port, frames are selected for transmission on the basis of the TCs that the port supports (D.21.1)
18 and the operation of the transmission selection algorithms supported by the corresponding queues on that
19 port. For a given port and traffic class, frames are selected from the corresponding queue for transmission if
20 and only if:

- 21 a) The operation of the transmission selection algorithm supported by that queue determines that there
22 is a frame available for transmission; AND
- 23 b) For each queue corresponding to a numerically higher value of traffic class supported by the port,
24 the operation of the transmission selection algorithm supported by that queue determines that there
25 is no frame available for transmission.

26 The order in which frames are selected for transmission from the queue maintains the ordering requirement
27 specified in [Q] 8.6.6.

28 **D.22 Transmission Gates ([Q] 8.6.8.4)**

29 Enhancements for Scheduled Traffic (EST) are supported according to Clause 8.

30 Use of EST is not required on any egress port where ATS or CBS are activated on any TC Queue.

31 NOTE—A rationale for not requiring ETS can be found in [B11] and 8.7.

32 **D.23 VID Translation ([Q] 6.9 f) and g))**

33 The VID Translation as specified by [Q] 6.9 f) and Egress VID translation as specified by [Q] 6.9 g) are
34 supported.

1 D.24 Frame Discard

2 Discarding a Frame is an intentional action by the Bridge Management Entity ([Q]8.12).

3 The bridge can discard a Frame:

- 4 a) Due to policing, if a flow metering algorithm ([Q] 8.6.5.5) determines that discard is necessary. ([Q]
5 6.5.2 b)7))
- 6 b) Due to policing, if a shaping algorithm ([Q] 8.6.11.3.7) determines that discard is necessary.
- 7 c) Due to policing, if the frame exceeds the queueMaxSDU ([Q] 8.6.8.4) for the TC queue. ([Q] 6.5.2
8 b)8))
- 9 d) Due to policing, if the max. SDU Size Filter ([Q] 8.6.5.3.1) determines that discard is necessary.
- 10 e) Due to congestion, if there is a risk of or actual exhaustion ([Q] 6.5.2 b) 2)) of internal buffer
11 capacity.
- 12 f) Due to an unknown egress port, if the FDB ([Q] 8.8) disallows the forwarding. ([Q] 6.5.2 b) 6))
- 13 g) Due to security considerations, if the device attached to the port is not authorized ([B5]) for access to
14 the network. ([Q] 6.5.2 b)5))

15 D.25 Frame Preemption

16 D.25.1 Preemptable MAC ([Q] 6.7.1 a))

17 The TCs queuing frames to the preemptable MAC do not include the numerically highest TC.

18 D.25.2 Express MAC ([Q] 6.7.1 b))

19 Frame preemption ([Q] 6.7.2) is configured as follows:

- 20 a) Use a single TC for queuing frames to the express MAC.
- 21 b) Have no shapers configured for the TC queuing frames to the express MAC.
- 22 c) Configure the TC queuing frames to the express MAC as the numerically highest.

23 D.26 Egress filtering ([Q] 8.6.4)

24 Frames are not transmitted on a port, if the member set ([Q] 8.8.10) for the frame's VID on that port is not
25 present.

26 D.27 ATS Scheduler Groups

27 All ATS Scheduler Instances processing Frames in a specific upstream TC arriving on one specific ingress
28 port belong to the same ATS scheduler Group.

29 If properly configured, and all stations operate according to their specification in the base standards
30 throughout the network, the order of Frames can not be distorted in the way described below.

31 If one ATS Scheduler Instance within an ATS scheduler Group has a significantly higher Committed
32 Information Rate, than a second ATS Scheduler Instance within the same ATS scheduler Group (ATS-
33 CIR[1] >> ATS-CIR[2]), the Group Eligibility Time shared between the two ATS Scheduler Instances leads

1 to a potentially undue delay of Frames if an ingress burst of Frames destined for the lower rate (ATS-
2 CIR[2]=r2) ATS Scheduler Instance within the Group pushes the Eligibility Time for the higher rate ATS
3 Scheduler Instance into the future. Worst case the Frame of the higher rate ATS Scheduler Instance might be
4 discarded, if $ATS-MRT[1] < L2/r2$, where $L2$ is the Frame's length.

5

1 Annex E

2 (informative)

3 Congestion

4 E.1 Purpose

5 This annex provides interpretations of normative features specified in the base standards as listed in
6 Clause 2, and guidance to usage of such features.

7 Implementations conformant to this standard solely need to conform to the corresponding normative
8 specification in the base standard that specifies a particular feature and is referred according to Clause 5 of
9 this standard. The interpretations described in this annex do not imply conformance requirements.

10 Any deviations imposed by this annex from the base standards in terms of additional definitions, limitations,
11 and alterations in this annex beyond the definitions in the base standards do not apply for conforming to the
12 base standards. Implementations solely following this annex instead of the definitions in the base standards
13 cannot claim conformance to the respective features from the base standards. Whenever a conflict between
14 this annex and the normative specification of features in the base standards arise, the normative specification
15 takes precedence over this annex.

16 E.2 Definitions

17 Congesting Traffic is providing excess ingress bandwidth, i.e. more Buffer is (or would be) required to store
18 Frames than was expected/configured.

19 Congested Traffic is under-served on egress, i.e. Frames accumulate in the Buffer of the Relay as they do not
20 get selected for transmission.

21 E.3 Congestion Separation

22 Congestion separation is to be performed on each bridge locally.

- 23 a) A bridge is configurable to segregate the Buffers for different Traffic, so Buffer overruns in one
24 segregated block can lead to discarding of Frames within that one Traffic aggregate, but not for the
25 other Traffic aggregates.
- 26 b) Frames are not forwarded on egress, if the bandwidth exceeds a configured maximum.
- 27 c) Frames are discarded if the bandwidth exceeds a configured maximum on ingress.

28 E.4 Causes for Congested Traffic

29 Traffic can become congested, if:

- 30 a) The Shaper (CBS, ATS, EST) on egress is configured with a too low rate or a too small burst size.
- 31 b) Excessively large lower priority Frames block the transmission.
- 32 c) A large higher Priority Burst blocks the transmission.

1 Therefore in order to prevent Traffic from becoming congested:

- 2 d) Bandwidth needs for all Traffic needs to be well known.
- 3 e) All traffic goes through a max. SDU Size Filter ([Q] 8.6.5.3.1).
- 4 f) All higher priority traffic (and therefore all traffic, except the lowest TC) goes through a Flow Meter
- 5 ([Q] 8.6.5.5).

6 **E.5 Causes for Congesting Traffic**

7 Traffic causes congestion, if:

- 8 a) Bursts of random size occur, as no shaping is implemented.
- 9 b) A Shaper (CBS, ATS, EST) with a mis-configured rate or burst size is active.
- 10 c) The Traffic contains excessively large Frames.

11 Therefore in order to prevent Traffic from causing congestion:

- 12 d) All Traffic is shaped on egress at every Talker.
- 13 e) Bandwidth needs for all Traffic need to be well known.
- 14 f) All traffic goes through a max. SDU Size Filter ([Q] 8.6.5.3.1).

15

1 Annex F

2 (informative)

3 Stream Identification

4 The Stream Identification of [CB] 6. differentiates active and passive stream identification functions. All
5 stream identification functions assign a *stream_handle* to the Frame they match in the input/up ([CB] 3.)
6 direction. The Active Destination MAC and VLAN Stream Identification of [CB] 6.6 modifies a frame in
7 the input/up as well as in the output/down ([CB] 3.) direction. Passive stream identification functions have
8 no effect on frames in the output/down direction.

9 The out-facing ([CB] 3.) passive stream identification functions serve to assign a *stream_handle* to the
10 Frames they match in the input/up direction to be used by the FRER functionality (Annex C.9) or the Stream
11 Filtering (D.16). As this standard does not make use of FRER, only the later usage is relevant. A passive in-
12 facing ([CB] 3.) stream identification function therefore can not serve any purpose in the context of this
13 standard, as the assigned *stream_handle* in the input/up direction could only be used by the FRER
14 functionality.

15 The Active Destination MAC and VLAN Stream Identification ([CB] 6.6) is used in the FRER context of
16 [CB] to allow frame replication of IP unicast Frames in the Relay by changing the original individual MAC
17 destination address to a group MAC address and thereby triggering egress to multiple ports. Due to its ISO/
18 OSI layer violating ([CB] 8.1 j)) nature and the ensued configuration complexity, this standard discourages
19 the use of active stream identification.

20 In conclusion there is no need to configure an in-facing stream identification function in the context of this
21 current standard.

22 NOTE 1—This description ignores the potential issues with the internal LAN ([CB] Figure 6-5) and the in-facing stream
23 identification functions removing the *stream_handle*, thereby preventing the Stream Filter (D.16) to identify Frames.

1 Annex G

2 (informative)

3 Non-real-time capable Protocols

4 Some protocols like TCP [B12] are intended to provide reliable data transfer, which in effect means, they
5 need to retransmit data, in case the receiver reports it as not having arrived within a certain time-frame.
6 These potential retransmissions makes the bandwidth demand for such TCP-flows unpredictable. An
7 unpredictable bandwidth presents a fundamental problem for configuring any of the TSN features described
8 in Clause , be it schedulers, shapers, or policers. In a scenario where frames are lost (e.g., due to congestion
9 or due to bit errors) the retransmission may trigger a policer to drop the unexpected higher bandwidth,
10 leading to further retransmissions, i.e., more bandwidth demand (see also C.7.1). Furthermore it becomes
11 difficult for such flows to configure a EST cycle fulfilling the stability condition described in 8.7.1.

12 As is discussed in E.5 the most rigorous policing is happening in the numerically highest TCs, as they can
13 have the most deteriorating effect on congestion. It is therefore not considered advisable to place TCP-flows
14 in the numerically highest TCs.

15

1 Annex H

2 (informative)

3 Egress Shaping in End Stations

4 In a bridge the source of the egress traffic at one port is the ingress on all other ports. Data can arrive
5 simultaneously on any ingress port, but needs to be transmitted in a serial fashion on the egress port.
6 Queuing frames according to Priority into TCs and the use of Shapers allows to control the order in which
7 frames are transmitted, as well as their spacing.

8 For an end station the ultimate source of data are the Applications. The Applications are themselves
9 scheduled by the Operating System (OS). In some environments Applications deliver data to a Middleware
10 (MW), which may have mechanisms to pack different data elements into various Frames, so one Frame does
11 not just carry the data from one Application. This obviously requires some data collection and potential
12 trigger conditions on how many of the planned Applications need to have contributed to the payload of a
13 Frame before it is actually transmitted by the Middleware. Therefore OS and MW have the capability to
14 shape the transmitted traffic.

15 In case a Middleware is not present, some OSs like Linux offer shaping mechanisms in software (SW), like
16 the Hierarchical Token Bucket (HTB). Also CBS like mechanisms are available in SW. Since SW on a
17 Talker can have more information available for its configuration, it can be much more flexible in assigning
18 Frames to certain shaper instances.

19 The shaping mechanisms are not very different. It is mainly the granularity at which Frames can be assigned
20 to Shapers and the number of Shaper Instances available, that actually make a difference in the overall
21 egress behavior.

22

Annex I

(informative)

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